

080103241030

Enrollment No. \_\_\_\_\_

1051

**APRIL-MAY 2025 EXAMINATION**  
**I B.Tech. (4YDC) EXAMINATION**  
**CO 10504/ CO 10507/: PROGRAMMING FOR PROBLEM SOLVING**

Time: 3 Hrs.]

[Max. Marks: 70

**TOTAL NO. OF QUESTIONS IN THIS PAPER: 5**

**NOTE: 1) Answer to the point and make assumptions wherever required and clearly state them.**  
**2) Attempt all parts of the same question at one place and each question carries five sub parts a, b, c, d, e. Part a, b, c is compulsory and attempt any one from d and e.**

| Ques. No. | Question                                                                                                                                                                                                                                                                                                                    | Marks | CO | BT | PI    |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|-------|
| Q.1 (a)   | Differentiate between application and system software with an example of each software.                                                                                                                                                                                                                                     | 02    | 1  | 2  | 2.1.1 |
| (b)       | Write a C statement using the ternary operator to assign grade 'A' if marks $\geq 80$ , 'B' if marks $\geq 60$ , and 'C' if marks $< 60$ .                                                                                                                                                                                  | 02    | 1  | 3  | 2.1.3 |
| (c)       | Compare primary, secondary, and cache memory in terms of speed, cost, and usage.                                                                                                                                                                                                                                            | 03    | 2  | 3  | 2.1.3 |
| (d) (i)   | Draw a flowchart to solve a quadratic equation by handling the cases for real and distinct roots, real and equal roots, and imaginary roots. Also, include a condition to handle a linear equation when the leading coefficient is zero.                                                                                    | 04    | 1  | 3  | 1.3.2 |
| (ii)      | Explain the difference between an error and a warning in a program. Can a program run with warnings but without errors?                                                                                                                                                                                                     | 03    | 2  | 2  | 1.2.3 |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                             |       |    |    |       |
| (e) (i)   | Draw a flowchart to take four points from user (x1,y1), (x2,y2), (x3,y3) and (x4,y4) and find out if these points form a parallelogram or not.                                                                                                                                                                              | 04    | 1  | 3  | 1.3.2 |
| (ii)      | Define a variable in C programming with an example. Explain how a variable is different from a keyword and an identifier in C.                                                                                                                                                                                              | 03    | 2  | 2  | 1.2.3 |
| Q.2 (a)   | What are the various primitive data types in C? List each type with an example and size. Assume a 64-bit system using GCC compiler.                                                                                                                                                                                         | 02    | 2  | 2  | 1.2.1 |
| (b)       | Explain type conversion and type casting with an example.                                                                                                                                                                                                                                                                   | 02    | 2  | 2  | 1.2.3 |
| (c)       | Write a C program to check if a number is divisible by 3, 5, or both and print corresponding messages: "Divisible by both 3 and 5", "Divisible by 3", "Divisible by 5", or "Not divisible by both 3 or 5".                                                                                                                  | 03    | 2  | 3  | 1.3.1 |
| (d)       | Evaluate the following expressions. Explain your answer with proper justification with steps.<br>int x = 4, y = 6, z = 3;<br>float a = 2.0, b = 3.5, c = 1.0, k, p, n, m;<br>k = x * y + (z <= a && b > c);<br>p = (x != y)    (a == b) && (z >= c);<br>m = (x > y && z < a)    b != c;<br>n = y / (z * x) > a ? 1.5 : 2.5; | 07    | 2  | 4  | 2.2.1 |

OR

- (e) Write a C program using a switch case to implement a bank system with options to check balance, deposit money, withdraw money, and exit. Ensure the program validates that deposit and withdrawal amounts are positive and that withdrawals do not exceed the balance. 07 2, 3 3
- Q.3 (a) Differentiate between while loop and do- while loop. 02 3 2
- (b) Differentiate between actual and formal parameters of the function. 02 3 2
- (c) Write a C program that takes a n x n matrix as input and prints the main diagonal and anti-diagonal elements using loops. 03 3 3
- Input:  
[a b c]  
[d e f]  
[g h i]  
Output:  
Main Diagonal = a, e, i  
Anti-Diagonal = c, e, g
- (d) (i) Given an array of integers and a target sum, write a recursive function to determine whether a subset exists whose sum is exactly equal to the target. 05 3 3
- Example: arr = {3, 34, 4, 12, 5, 2}, target = 9 → Output: True  
(Because 4 + 5 = 9)
- (ii) How does a recursive function differ from an iterative function? 02 3 2
- OR
- (e) (i) Write a C function int nextBinaryPalindrome(int K) that returns the smallest integer greater than K whose binary representation is a palindrome. 05 3 3
- Input: A positive integer K  
Output: The smallest binary palindrome strictly greater than K
- (ii) What is scope in the context of functions in C? 02 3 2
- Q.4 (a) Define a character array in C. How is it different from a string in C? 02 4 2
- (b) Given the following string declarations: 02 3 2
- ```
char str1[20] = "Hi";
char str2[20] = "Bye";
```
- Write the output of each of the following string functions:
- ```
strcpy(str1, str2);
strcat(str1, "!!!");
strlen(str1);
strcmp(str1, str2);
```



- (c) Given an integer array, write a C program to find the index  $i$  such that the sum of all elements before index  $i$  is equal to the sum of all elements after it.  
Print the index if it exists; otherwise, print "No such index found."  
Example: Input: {1, 4, 1 5, 3, 3} → Output: Index 3 03 3 3 4.5.1
- (d) (i) Define a structure in C. How is it different from an array? Explain with example. 03 4 2 1.4.1
- (ii) Define a structure Book with members: title, author, and price. Write a C program to input details of 3 books and display them. 04 4 3 1.5.2

OR

- (e) (i) What is an array in C? Compare one-dimensional and multi-dimensional arrays. How are both declared? 03 3 2 4.5.1
- (ii) Write a program to remove all duplicate characters from a given string without changing their original order.  
Example: "programming" → Output: "progamin" 04 3 3 4.5.2

- Q.5 (a) Explain the role of the & (address-of) and \* (dereference) operators in pointers with examples. 02 4 1 2.4.1
- (b) Write a C program to open a file in write mode, write some text into it, and then close the file. 02 4 2 4.5.2
- (c) Write a program to reverse a string using pointers. 03 4 3 4.5.2
- (d) (i) Compare object-oriented programming and procedure-oriented programming with examples. 04 4 2 1.2.1
- (ii) Create a base class Shape with a function displayArea(). Derive two classes, Circle and Rectangle, from Shape. Implement the displayArea() function in both derived classes to calculate and display the area of the respective shapes. Write a C++ program to display the area of both shapes. 03 4 4 3.1

OR

- (e) (i) Explain the given OOPS concepts: inheritance, polymorphism and abstraction. 04 4 2 1.2.1
- (ii) Create a base class Vehicle with data members brand, model, and year. Derive a class Car from Vehicle that adds attributes seatingCapacity and fuelType. Write a C++ program to display the details of both the Vehicle and Car classes. 03 4 3 4.3.2

XXXXX

**DEC 2024 EXAMINATION**  
**I B.Tech. (4YDC) EXAMINATION**  
**CO10504/ CO10507/: PROGRAMMING FOR PROBLEM SOLVING**

Time: 3 Hrs.]

[Max. Marks: 70]

**TOTAL NO. OF QUESTIONS IN THIS PAPER: 5**

**NOTE:** 1) Answer to the point and make assumptions wherever required and clearly state them.  
 2) Attempt all parts of same question at one place and each question carries five sub parts a, b, c, d, e. Part a, b, c is compulsory and attempt any one from d and e.

|           | Question                                                                                                                                                                                                                                                                                                                                                        | Marks | CO  | BT  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| Q.1 (a)   | Differentiate between:<br>i) Application and System Software<br>ii) Keywords and Identifiers                                                                                                                                                                                                                                                                    | 02    | CO1 | BT2 |
| (b)       | Identify the most appropriate answer of the following :<br>i) <u>cp</u> command is used to copy a file in linux environment.<br>ii) Computers use the <u>binary</u> language to process data.<br>iii) Monitor and Printer are examples of _____ types of devices.<br>iv) <u>hard disk</u> is a type of memory that is non-volatile and stores data permanently. | 02    | CO1 | BT1 |
| (c)       | Answer the following:<br>(i) How many bits are there in the following: 2 byte, 2 megabyte, and 2 gigabyte?<br>(ii) Explain the structure of C program with an example.                                                                                                                                                                                          | 03    | CO2 | BT2 |
| (d) (i)   | Draw a flowchart to validate a given date (day, month, and year), including checking if the year is a leap year for February, and ensuring the day is within valid ranges for all months.                                                                                                                                                                       | 05    | CO1 | BT1 |
| (ii)      | What is the role of the compiler in detecting programming errors?                                                                                                                                                                                                                                                                                               | 02    | CO2 | BT1 |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                 |       |     |     |
| (e) (i)   | Draw a flowchart that takes the coordinates of three points (x1, y1), (x2, y2), and (x3, y3) and checks if they form a triangle, considering the collinearity of the points.                                                                                                                                                                                    | 05    | CO1 | BT1 |
| (ii)      | Define a variable in programming. What is the difference between a variable and a constant in C?                                                                                                                                                                                                                                                                | 02    | CO2 | BT1 |
| Q.2 (a)   | List the different types of data types in C. How its range is calculated.                                                                                                                                                                                                                                                                                       | 02    | CO2 | BT1 |
| (b)       | Explain the concept of type promotion and type conversion in C. Provide an example of both during an arithmetic operation.                                                                                                                                                                                                                                      | 02    | CO2 | BT1 |
| (c)       | Write a C program using if-else that takes a 4-digit integer as input and finds the sum of the 1st and 3rd digit and 2 <sup>nd</sup> and 4 <sup>th</sup> digit.                                                                                                                                                                                                 | 03    | CO2 | BT1 |
| (d)       | Evaluate the expressions to find the output of the program. Explain your answer with proper justification with steps.<br>#include <stdio.h><br>int main(){                                                                                                                                                                                                      | 07    | CO2 | BT1 |



```

int x = 2, y = 3, z = 5, n;
float a = 1.5, b = 2.5, c = 3.5, k, p, m;
k = (int)b * (y % x) * y / z / c;
p = x++ / y * z-- + b / a * c;
m = y < c || x >= a * ++y && z != 5 * b;
n = x % z + a - c * b / z;
printf("k=%f p=%f m=%f n=%d", k, p, m, n);
return 0;

```

OR

- (e) Write a C program using switch statement that takes annual income as input and calculates the income tax based on the following income tax slabs: 07 CO2, BT4 2.5.2 CO3

- Income  $\leq$  Rs. 10,000: No tax
- Income Rs. 10,001–20,000: 10% on income exceeding Rs. 10,000
- Income Rs. 20,001–30,000: 15% on income exceeding Rs. 20,000, plus 10% on Rs. 10,000 (Rs. 10,001–20,000)
- Income Rs. 30,001–40,000: 20% on income exceeding Rs. 30,000, plus 15% on Rs. 10,000 (Rs. 20,001–30,000), and 10% on Rs. 10,000 (Rs. 10,001–20,000)
- Income Rs. 40,001–50,000: 25% on income exceeding Rs. 40,000, plus 20% on Rs. 10,000 (Rs. 30,001–40,000), 15% on Rs. 10,000 (Rs. 20,001–30,000), and 10% on Rs. 10,000 (Rs. 10,001–20,000)
- Income  $>$  Rs. 50,000: 30% on income exceeding Rs. 50,000, plus taxes for all previous slabs.

- Q.3 (a) Write a code to differentiate between while and do-while loop. 02 CO3 BT3 1.7.1
- (b) Write a code to differentiate call by value and call by reference. 02 CO3 BT3 2.5.2
- (c) Write a function in C to find one pair of prime numbers in a given range such that the difference between them is also a prime number. The program should stop after finding the first valid pair. 03 CO3 BT4 3.8.2
- (d) Write a recursive program to calculate the value of  $nCr$ , which represents the number of ways to choose  $r$  items from a set of  $n$  items. The recurrence for calculating  $nCr$  is given by:  $nCr = {}^{n-1}C_r + {}^{n-1}C_{r-1}$ . Assume that  $n \geq r$ . 07 CO3 BT4 3.8.2

OR

- (e) Write a C program with a function to rotate an array of  $n$  elements to the left by  $d$  positions in-place. The rotation should modify the array directly, without using any additional arrays. 07 CO3 BT4 3.8.2
- Input: Array:  $arr[] = \{1, 2, 3, 4, 5\}$ ,  $d = 2$
- Output: Rotated array:  $\{3, 4, 5, 1, 2\}$

- Q.4 (a) Differentiate between `strlen`, `strcpy`, `strcat`, and `strcmp`. 02 CO4 BT2 2.5.2
- (b) Write a C program to count the number of vowels in a string using pointers. 02 CO4 BT3 1.7.1
- (c) An element of the array is called a leader if it is greater than all the elements to its right. Write a program in C to find all the "leader" elements in a given array. 03 CO3 BT4 4.5.1

- (d) Define multidimensional array in C. How can an array be initialized in C. Write a program in C to find the saddle point of a matrix. A saddle point is an element which is the minimum element in its row but the maximum element in its column.

07 CO3 BT4 4

OR

- (e) What is a structure in C? How do you declare and define a structure in C? Write a C program to define a structure Employee with the attributes: Name, Age, Salary. Create an array of 3 employees, take their details from the user, and display the details of employees earning a salary greater than 50,000 using structures.

07 CO4 BT4 4

- Q.5 (a) What is the difference between `*ptr++` and `(*ptr)++`? Explain with an example

02 CO4 BT2 1

- (b) What is dynamic memory allocation in C? List functions that are used for dynamic memory allocation in C?

02 CO4 BT2 1

- (c) Explain the purpose of the following file functions: `fgetc()`, `fputc()`, `fgets()`, `fputs()`, `fprintf()`, `fscanf()`

03 CO4 BT2 1

- (d) What are abstract classes? How are they implemented in C++? Write a program to create a class Student with data members name, roll number, and marks. Include methods for setting and displaying data.

07 CO4 BT4 4

OR

- (e) Define inheritance in C++. Explain its importance in Object-Oriented Programming. Write a C++ program to demonstrate single inheritance. Create a base class Person with attributes name and age and a method to display them. Derive a class Student from Person with an additional attribute marks and a method to display all the details.

07 CO4 BT4 4

XXXXX



May-June 2024 EXAMINATION  
I B.Tech. (4YDC) EXAMINATION  
CO10504/ CO10507: PROGRAMMING FOR PRBLEM SOLVING

3 Hrs.]

[Max. Marks: 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

1) Answer to the point and make assumptions wherever required and clearly state them.  
pt all parts of same question at one place and each question carries five sub parts a, b, c, d, e. Part a, b, c, d, e are compulsory and attempt any one from d and e.

|    | Question                                                                                                                                                                                                                                                                                                    | Marks | CO  | BT  | PI    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-------|
| a) | i) _____ operator returns the number of bytes occupied by the operand.<br>ii) The _____ specification is used to read/write a short integer.<br>iii) The _____ statement is used to skip statements in a loop.<br>iv) MS Word is an example of _____ software.                                              | 02    | CO1 | BT1 | 1.7.1 |
| b) | What is preprocessor directive? Explain #define and #include preprocessor directives                                                                                                                                                                                                                        | 02    | CO1 | BT1 | 1.7.1 |
| c) | Evaluate the expressions for the following, assume that the variables are declared and initialized as follows:<br>int x = 3, y = 4, z = 5;<br>float a = 2.5, b = 3.5, c = 4.5;<br>i) (float)b * (y % x) * y / 2    z * x;<br>ii) ++x / 4 * z++ + y / x++ * b;<br>iii) y < c    x > a * ++x && z <= 5 * b;   | 03    | CO1 | BT2 | 1.7.1 |
| d) | (i) Take three points as input (x1, y1), (x2, y2), and (x3, y3). Draw a flow chart to find whether the points form a triangle or not.                                                                                                                                                                       | 05    | CO1 | BT3 | 2.5.2 |
|    | (ii) How run-time error is different from compile-time errors?                                                                                                                                                                                                                                              | 02    | CO1 | BT2 | 1.7.1 |
|    | <b>OR</b>                                                                                                                                                                                                                                                                                                   |       |     |     |       |
| e) | (i) Draw a flowchart to calculate the sum of given series:<br>$x^1 + x^3 + x^5 + x^7 + \dots$ upto n terms.                                                                                                                                                                                                 | 05    | CO1 | BT3 | 2.5.2 |
|    | (ii) State True or False<br>1. printf("%d", scanf("%d", &num)); is a valid C statement.<br>2. X=Y=2.4; is valid C statement.<br>3. Logical errors are detected by the compiler.<br>4. Syntax error is generated when rules of programming language are violated.                                            | 02    | CO1 | BT2 | 1.7.1 |
| a) | Define type casting and type conversion with the help of an example.                                                                                                                                                                                                                                        | 02    | CO2 | BT2 | 1.7.1 |
| b) | For the following while loop, write the equivalent do while loop statements<br><pre>#include &lt;stdio.h&gt; #include &lt;stdlib.h&gt; int main() {     int i=1, j, k, m2=10, num=0;     printf("Print the information\n");     while(i&lt;=m2)     {         int k=1;         while(k&lt;=m2-(m2-i))</pre> | 02    | CO2 | BT2 | 1.7.1 |

|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|     |     | <pre> {     printf(" ");     k++; } num=m2-i; j=1; while(j&lt;=num) {     printf("%d",m2-(m2-j));     j++; } i++; printf("\n"); } return 0; } </pre>                                                                                                                                                                                                                                                                                                                  |    |
|     | (c) | <p>Write a C program to read an amount (integer value) and break the amount into the smallest possible number of bank notes. (Currency notes are available as:(100,50,20,10,5,2,1))</p> <p>Test Data :</p> <p>Input the amount: 375</p> <p>Expected Output:</p> <p>There are:</p> <p>3 Note(s) of 100.00</p> <p>1 Note(s) of 50.00</p> <p>1 Note(s) of 20.00</p> <p>0 Note(s) of 10.00</p> <p>1 Note(s) of 5.00</p> <p>0 Note(s) of 2.00</p> <p>0 Note(s) of 1.00</p> | 03 |
|     | (d) | <p>Write a function in C to print following pattern</p> <p>Note: Only number of columns will be the input parameter to the Function</p> <p>Mirrored Parallelogram</p> <pre> ***** ***** ***** ***** </pre>                                                                                                                                                                                                                                                            | 07 |
|     |     | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|     | (e) | <p>Write a program to find the difference of two dates in years, month, and days. Find whether the entered dates are valid or not. Assume that the first date falls before the second date. (Date format is DD/MM/YYYY)</p>                                                                                                                                                                                                                                           | 07 |
| Q.3 | (a) | What is a null statement? How can it be useful in our programs?                                                                                                                                                                                                                                                                                                                                                                                                       | 01 |
|     | (b) | <p>What will be the output of the following program, when program is executed? Justify your answer.</p> <pre> #include&lt;stdio.h&gt; int main() {     int i, sum1 = 0, sum2 = 0;     for ( i = 1; i&lt;=11 ; i++)         if((i*i*i*2+3&lt;=(i*i*4+5))             sum1++;         else             sum2++; } </pre>                                                                                                                                                 | 01 |



|        |                                 | printf(“%d %d” ,sum1, sum2);<br>return 0;<br>}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |             |        |                  |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
|--------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------|--------|------------------|----|---------------------------------|----|---|----|---------------------------------|----|---|----|---------------------------------|---|---|----|-----|-----|-------|
| (c)    |                                 | What is Recursion? Write a C program to compute polynomial co-efficient nCr using recursion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 03               | CO3         | BT4    | 3.8.2            |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| (d)    |                                 | Write a C program that takes three coefficients (a,b,and c) of a quadratic equation ; (ax <sup>2</sup> +bx+c) as input and compute all possible roots and print them with appropriate messages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 07               | CO3         | BT3    | 3.8.2            |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
|        |                                 | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |             |        |                  |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| (e)    |                                 | <p>A prime number is one that can only be factored by itself or one. A number is called a happy number, if you repeat the process, of squaring the sum of the digits, till the value 1 is obtained. A happy prime is a number that is happy and prime. Write a program that checks whether the number entered by the user is a happyprime number or not for 10 cycles/iteration only.<br/>Example: Assume a number ‘23’<br/>Number Computation Result Cycles/Iteration</p> <table><tr><th>Number</th><th>Computation</th><th>Result</th><th>Cycles/Iteration</th></tr><tr><td>23</td><td>3<sup>2</sup> + 2<sup>2</sup></td><td>13</td><td>1</td></tr><tr><td>13</td><td>3<sup>2</sup> + 1<sup>2</sup></td><td>10</td><td>2</td></tr><tr><td>10</td><td>0<sup>2</sup> + 1<sup>2</sup></td><td>1</td><td>3</td></tr></table> <p>So number ‘23’ is a happy prime number.</p> | Number           | Computation | Result | Cycles/Iteration | 23 | 3 <sup>2</sup> + 2 <sup>2</sup> | 13 | 1 | 13 | 3 <sup>2</sup> + 1 <sup>2</sup> | 10 | 2 | 10 | 0 <sup>2</sup> + 1 <sup>2</sup> | 1 | 3 | 07 | CO3 | BT3 | 3.8.2 |
| Number | Computation                     | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Cycles/Iteration |             |        |                  |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| 23     | 3 <sup>2</sup> + 2 <sup>2</sup> | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                |             |        |                  |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| 13     | 3 <sup>2</sup> + 1 <sup>2</sup> | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                |             |        |                  |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| 10     | 0 <sup>2</sup> + 1 <sup>2</sup> | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                |             |        |                  |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| (a)    |                                 | <p>Explain the output of the following C code with proper justification?</p> <pre>#include &lt;stdio.h&gt; int main(){     int arr[5]= {1, 2, 3, 4, 5};     int *ptr1= arr;     int *ptr2= ptr1+3;     int *ptr3= ptr2+3;     printf(“%d\n”, *ptr2);     printf(“%d\n”, ptr2-ptr1);     printf(“%d\n”, *(arr+4));     return 0; }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 02               | CO4         | BT4    | 2.5.2            |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| (b)    |                                 | Can we have array of function pointers? if yes, illustrate with the help of suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 02               | CO4         | BT2    | 1.7.1            |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| (c)    |                                 | Write a C function isprime(num) that accepts an integer argument and returns 1 if the argument is a prime or 0 otherwise. Write a program that invokes this function to generate prime numbers between the given ranges                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 03               | CO4         | BT3    | 4.5.1            |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |
| (d)    |                                 | <p>Given an array of size N. Let A and B be the minimum and maximum in the array respectively. Task is to find how many number should be added to the given array such that all the element in the range [A, B] occur at-least once in the array.<br/>Examples:</p> <p>Input : arr[] = {4, 5, 3, 5, 6}<br/>Output : 1<br/>Only 7 to be added in the list</p> <p>Input : arr[] = {2, 1, 3}<br/>Output : 0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 07               | CO4         | BT3    | 4.5.1            |    |                                 |    |   |    |                                 |    |   |    |                                 |   |   |    |     |     |       |

|     |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |
|-----|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|     |     |  | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |
|     | (e) |  | Write a C program using function that implements a quiz consisting of a single question with four options, one of which is correct (use Function to display the quiz question and options). Upon user input, the program should determine and display whether the provided answer is correct or incorrect.<br>Expected Output:<br>What is the capital of Japan?<br>A. Beijing<br>B. Seoul<br>C. Tokyo<br>D. Bangkok<br>Your answer (enter A, B, C, or D): C<br>Correct! Well done. | 07   |
| Q.5 | (a) |  | What is the difference between the function malloc() and calloc()? Explain with example.                                                                                                                                                                                                                                                                                                                                                                                           | 02   |
|     | (b) |  | The following recursive function takes a string of given length as input and determines whether the string is a palindrome. It returns 0 if the string is not a palindrome and 1, if it is. Fill in the blanks. Each blank can have AT MOST one statement.<br>int ispalindrome(char A[ ], int n )<br>{<br>if ( _____ ) return 1;<br>if ( _____ ) return 0;<br>return ispalindrome(_____); }                                                                                        | 02   |
|     | (c) |  | Explain the visibility of base class members for the access specifiers: private, protected, and public.                                                                                                                                                                                                                                                                                                                                                                            | 03   |
|     | (d) |  | Differentiate between the following:<br>(i) Function overloading and Operator overloading<br>(ii) Abstraction and Encapsulation<br>(iii) Static and Dynamic binding                                                                                                                                                                                                                                                                                                                | 07   |
|     |     |  | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |
|     | (e) |  | Write a short note on the following functions. For each function, give a program code that demonstrates its usage.<br>(a) fopen()<br>(b) fclose()<br>(c) ferror()<br>(d) clearerr()<br>(e) fseek()                                                                                                                                                                                                                                                                                 | - 07 |
|     |     |  | XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |



**SGSITS-CAMPUS-PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,  
Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

311

DEC - 2023 EXAMINATION

I B.E. / B. Tech. (4YDC) EXAM

CO10504/CO10507: PROGRAMMING FOR PROBLEM SOLVING

Time: 3 Hrs.]

[Max. Marks: 70

[Min. Pass. Marks: 22

**TOTAL NO. OF QUESTIONS IN THIS PAPER : 5**

NOTE: Attempt all the questions. Each question have five sub-parts a, b, c, d and e. Sub-part a, b and c are compulsory and attempt any one part out of (d) and (e). Attempt all sub-parts of a question at one place.

- |                                                                                                                                                                                                                                                                                                                                                                              | Marks | CO  | BL  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| 1. (a) The following names are chosen by a programmer to be used as variable names in the program. Which of these is an illegal choice? Also, give the reason.<br>Myname_123, Case, Value_of_Pi_correct_upto_7_decimal_digits, do_while, intHeight, 2values and total%                                                                                                       | [2]   | CO1 | BL2 |
| (b) What are the different types of memory used by the computer? Also, write their purpose and use.                                                                                                                                                                                                                                                                          | [2]   | CO1 | BL1 |
| (c) What are the different types of special symbols supported by the C programming language in pairs? Also enlist their use and role in the programming.                                                                                                                                                                                                                     | [3]   | CO2 | BL1 |
| (d) Draw the flowchart to check whether the number entered by the user is a prime palindrome or not. For example, 131 is a prime palindrome number because it's a prime number and also a palindrome number.                                                                                                                                                                 | [7]   | CO1 | BL3 |
| <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                    |       |     |     |
| (e) Draw the flowchart to convert a decimal number entered by the user into its equivalent binary and octal representation.                                                                                                                                                                                                                                                  | [7]   | CO1 | BL3 |
| 2. (a) What is the use of the default case in a switch statement? Is a default case necessary in a switch statement?                                                                                                                                                                                                                                                         | [2]   | CO2 | BL2 |
| (b) Write only if-else statement to determine whether an entered character (char ch) is a vowel (in both upper and lower cases) or not a vowel.                                                                                                                                                                                                                              | [2]   | CO2 | BL3 |
| (c) When following compilation errors will occur when you compile the program.<br>(i) expected ';' before 'scanf'<br>(ii) expected declaration or statement at end of input<br>(iii) undefined reference to 'main'                                                                                                                                                           | [3]   | CO1 | BL2 |
| (d) What is output of the following printf() statements:<br>i) printf("%d", 7 << 2   3);<br>ii) printf("%d", printf("%d", 1 << 10));<br>iii) printf("%f", 1/2 * 10 * 20);<br>iv) printf("%d", 2 << 3 >> 2);<br>v) printf("%d", (6 % 3) ? 4 & 6 : 4   6);<br>vi) printf("%d", 8/7 * 6/5 * 4/3 * 2 + 1 - 8);<br>vii) printf("%d", 3 - 3 * 2    4 % 2 * 3 & & 4.0 / 5 * 2 - 3); | [7]   | CO3 | BL2 |
| <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                    |       |     |     |
| (e) i) For the C expression (((- 2) * (x - 3)) < (x + (2 * z))) && (z < 10), remove all the unnecessary parenthesis so that the meaning of the statement does not change.<br>ii) What are type conversion and type casting? Explain with some examples.<br>iii) Write a program for showing the use of if-else with a break, continue, and exit.                             | [7]   | CO2 | BL2 |
| 3. (a) The two expressions ('0' == ('1' - 1)) and (12 == (6 + '6')) will generate the same value. Justify your answer.                                                                                                                                                                                                                                                       | [2]   | CO2 | BL2 |
| (b) What is recursion? Differentiate between recursion and iteration.                                                                                                                                                                                                                                                                                                        | [2]   | CO3 | BL1 |
| (c) How many different types of looping statements are there in C programming? Explain the use of each loop with some real-life programming examples.                                                                                                                                                                                                                        | [3]   | CO3 | BL1 |
| (d) For given integer numbers a and b, write a program to print all palindromic prime numbers that are smaller than or equal to b and greater than or equal to a.                                                                                                                                                                                                            | [7]   | CO3 | BL3 |

**OR**

**SGSITS CAMPUS PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

- (e) A special number is a number in which the sum of the factorial of each digit is equal to the number itself. For example:  $145 = 1! + 4! + 5! = 1 + 24 + 120$ .  
Write a program to check, if a given number is a special number or not. [7]

- Q 4. (a) If we have a character array `char rtr[10] = "CO10504"`; then the expressions `strlen(rtr)` and `sizeof(rtr)` will give us the same value. Justify your answer. [2]  
(b) How do you find the address of an elements in 2D array? If a 2D array is arranged in column-major ordering. [2]  
(c) Write a program to add two complex number using structure or union. [3]  
(d) Write a program to rotate a given square matrix of size  $N \times N$  by 45 degrees clockwise. Print both the original and rotated matrices. [7]

For Example: if  $N=3$ 

Original Matrix:

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

Rotated Matrix:

|   |   |   |
|---|---|---|
|   | 1 |   |
| 4 |   | 2 |
| 7 | 5 | 3 |
|   | 8 | 6 |
|   | 9 |   |

OR

- (e) Write a program to rotate a given square matrix of size  $N \times N$  by 45 degrees anti-clockwise. Print both the original and rotated matrices. [7]

For Example: if  $N=3$ 

Original Matrix:

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

Rotated Matrix:

|   |   |   |
|---|---|---|
|   |   | 3 |
|   | 2 | 6 |
| 1 | 5 | 9 |
|   | 4 | 8 |
|   | 7 |   |

- Q 5. (a) Identified which concept of OOPS involved in below scenarios: [2]  
(i) Dialing a number call from some mobile device that call some method internally which concatenate the numbers and displays it on screen but what is it doing we don't know.  
(ii) Bluetooth which we usually have it in our mobile. When we switch on a Bluetooth, I am able to connect to another mobile or Bluetooth enabled devices but I'm not able to access the other mobile features like dialing a number, accessing inbox etc.  
(b) Enlist any two benefits of Object oriented methodology. [2]  
(c) What do you understand by identity, state and behavior of object? Explain with reference to real world example. [3]  
(d) The university conducts many courses and students can register for those courses. The registrar of the university can introduce any new course. Also the registrar maintains the curriculum and students details. The examination are conducted for the students and a degree is issued for the students who have completed the courses successfully. [7]  
Analyze the given case study, identify and enlist the classes, objects, attributes and behaviors and OOP concept used. State reasons for your choice.

OR

- (e) Create a class `Animal` which holds attributes `name` and `noOfLegs` and has a method `makeSound()`, which should just output "This animal makes sound". Derive child class from this `Animal` class `Dog(barks)`, `Cat(meows)`, and `Horse(neighs)`. Some other `Animals` have also been created, but some of them do not make any special sound. [7]  
Write a main function that prints sound of every animal that is passed to it as an array of animals.

\*\*\*\*\*



**SGSITS-CAMPUS-PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

1072

JUNE 2023 EXAMINATION

I B.Tech. (4YDC) EXAMINATION

CO10504/ CO10507: PROGRAMMING FOR PROBLEM SOLVING

Time: 3 Hrs.]

[Max. Marks: 70

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

E: 1) Answer to the point and make assumptions wherever required and clearly state them.  
Attempt all parts of same question at one place and each question carries five sub parts a, b, c, d, e. Part a, b, c are compulsory and attempt any one from d and e.

|     | Question                                                                                                                                                                                                                                                                                                                                                                           | Marks | CO  | BT  | PI    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-------|
| (a) | Differentiate between:<br>(i) RAM and ROM<br>(ii) Keywords and Identifiers                                                                                                                                                                                                                                                                                                         | 02    | CO1 | BT1 | 1.7.1 |
| (b) | Write the difference between type casting and type conversion with example.                                                                                                                                                                                                                                                                                                        | 02    | CO1 | BT1 | 1.7.1 |
| (c) | What is Variable? What are the rules to construct identifier (variable)? Classify the following as valid/invalid Identifiers.<br>i) num2 ii) \$num1 iii) +add iv) a_2 v) 199_space vi) _apple vii) #12                                                                                                                                                                             | 03    | CO1 | BT2 | 1.7.1 |
| (d) | Answer the following:<br>(i) Explain the steps involved in solving a problem using computer.<br>(ii) What is the difference between creating constant using #define macro and const keywords?<br>(iii) Write 3 examples of input and output devices.                                                                                                                               | 07    | CO1 | BT3 | 2.5.2 |
| OR  |                                                                                                                                                                                                                                                                                                                                                                                    |       |     |     |       |
| (e) | Define Algorithm. Write an algorithm and draw a flowchart to find the area and perimeter of a circle:                                                                                                                                                                                                                                                                              | 07    | CO1 | BT3 | 2.5.2 |
| (a) | What are enumerated data types? Explain with example.                                                                                                                                                                                                                                                                                                                              | 02    | CO2 | BT2 | 1.7.1 |
| (b) | Evaluate the following expressions:<br>i) $22 + 3 < 6 \ \&\& \ 15 \parallel 22 == 7 \ \&\& \ 22 - 2 > +5$<br>ii) $a + 2 > b \parallel !c \ \&\& \ a == d * a - 2 < = e$ Where $a=11, b=6, c=0, d=7$ and $e=5$ .                                                                                                                                                                    | 02    | CO2 | BT2 | 1.7.1 |
| (c) | Write the if-else-if equivalent code for the following code snippet:<br><pre>#include&lt;stdio.h&gt; main() {     char grades;     scanf("%c", &amp;grades);     switch(grades)     {         case 'A': { printf("Excellent"); break; }         case 'B': { printf("Very Good"); break; }         case 'C': { printf("Good"); }         default: { printf("Poor"); }     } }</pre> | 03    | CO2 | BT2 | 1.7.1 |
| (d) | Write a C program that takes three coefficients (a,b, and c) of a quadratic equation $ax^2 + bx + c$ as input and compute all possible roots and print them with appropriate messages.                                                                                                                                                                                             | 07    | CO2 | BT3 | 1.2.1 |
| OR  |                                                                                                                                                                                                                                                                                                                                                                                    |       |     |     |       |
| (e) | Write a program in C using functions to swap two numbers using global variables concept and call by reference concept.                                                                                                                                                                                                                                                             | 07    | CO4 | BT3 | 2.5.2 |
| (a) | What is the difference between the following declarations?<br><code>int arr1[8]; int (*arr2)[8]; int *(arr3[8]);</code>                                                                                                                                                                                                                                                            | 02    | CO3 | BT2 | 1.7.1 |

# SGSITS-CAMPUS PHOTOCOPY CENTRE

Below Gym Centre, SGSITS Campus

Mobile : 9893555950

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |     |     |
|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|
|     | (b) | Consider a positive integer n which can be expressed as a power of 4. Fill up the dashed lines in the program.<br>#include <stdio.h><br>int main()<br>{<br>int n, temp, flag;<br>scanf ("%d",&n);<br>temp=n;<br>flag=0;<br>while (____) {<br>if(____) {<br>flag=1;<br>break; }<br>temp=____; }<br>if(____)<br>printf("%d is a number that can be expressed as power of 4",n);<br>else<br>printf("%d cannot be expressed as power of 4", n);<br>} | 02 | CO3 | BT  |
|     | (c) | Write the following function in a recursive manner.<br>int count=0, s=2;<br>void sum_A(int x) {<br>while(x>count) {<br>s *= x;<br>x++;<br>count++;<br>} }                                                                                                                                                                                                                                                                                        | 03 | CO3 | BT  |
|     | (d) | Define Function and Write a function/program that takes an array of strings (char [ ][ ] or *char[]) and prints them, one per line, in a rectangular frame. For example the list ["Hello", "World", "of", "c", "print"] gets printed as:<br>*****<br>* Hello *<br>* World *<br>* of *<br>* c *<br>* print *<br>*****                                                                                                                             | 07 | CO3 | BT  |
|     |     | OR                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |     |     |
|     | (e) | Define structure and write a program using structure data type called TimeStruct containing 3 members called hour, minute and second. Develop a program that would assign values to the individual members and display the time in the form 16:40:30.                                                                                                                                                                                            | 07 | CO3 | BT3 |
| Q.4 | (a) | Give any 2 differences between while statement and a Do- while statement in C.                                                                                                                                                                                                                                                                                                                                                                   | 02 | CO3 | BT4 |
|     | (b) | Show how break and continue statements are used in a C-program, with example.                                                                                                                                                                                                                                                                                                                                                                    | 02 | CO4 | BT2 |
|     | (c) | Write a program to reverse a string. For example:<br>Input String: hello.<br>Output string: olleh                                                                                                                                                                                                                                                                                                                                                | 03 | CO4 | BT3 |
|     | (d) | Write a program to multiply two arrays of integers.<br>For Example Input: a[] = {1, 9, 3, 4, 5}, b[] = {6, 7, 8}.<br>Output: c[] = {1, 3, 1, 1, 5, 9, 1, 0} (i.e. 19345 * 678 = 13115910)                                                                                                                                                                                                                                                        | 07 | CO4 | BT3 |
|     |     | OR                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |     |     |
|     | (e) | Given a text file, write a program to create another text file deleting the words "a", "the", "an" and replacing each one of them with a blank space.                                                                                                                                                                                                                                                                                            | 07 | CO4 | BT3 |
| Q.5 | (a) | Enlist the major differences between procedures oriented and object oriented                                                                                                                                                                                                                                                                                                                                                                     | 02 | CO4 | BT2 |



**SGSITS-CAMPUS-PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

|     |                                                                                                                                                                                                  |    |     |     |       |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|-------|
| (b) | Differentiate between compile time and run time polymorphism.                                                                                                                                    | 02 | CO4 | BT2 | 1.7.1 |
| (c) | Explain the visibility of base class members for the access specifiers: private, protected, and public.                                                                                          | 03 | CO4 | BT2 | 1.7.1 |
| (d) | Differentiate between the following with programming example:<br>i) Function overloading and Operator overloading<br>ii) Object and Class<br>iii) Abstraction and Encapsulation                  | 07 | CO4 | BT3 | 4.5.1 |
|     | OR                                                                                                                                                                                               |    |     |     |       |
| (e) | Differentiate between the following with programming example:<br>i) Overloading and overriding<br>ii) Multiple Inheritance and multilevel inheritance<br>iii) Static binding and dynamic binding | 07 | CO4 | BT3 | 4.5.1 |
|     | XXXXX                                                                                                                                                                                            |    |     |     |       |

**SGSITS-CAMPUS PHOTOCOPY CENTRE**

Below Gym Centre, SGSITS Campus

Mobile : 9893555950

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

FEB 2023 EXAMINATION

I B.Tech. (4YDC) EXAMINATION

CO10504/ CO10507: PROGRAMMING FOR PROBLEM SOLVING

Time: 3 Hrs.]

[Max. Marks]

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

NOTE: 1) Answer to the point and make assumptions wherever required and clearly state them.

2) Attempt all parts of same question at one place and each question carries five sub parts a, b, c, d, e. c is compulsory and attempt any one from d and e.

| Question No. | Question                                                                                                                                                                                                                                                                                                                                                                                          | Marks | CO |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|
| Q.1 (a)      | Differentiate between:<br>i) Source code file and executable code file<br>-ii) Keywords and Identifiers                                                                                                                                                                                                                                                                                           | 02    | CO |
| (b)          | Identify the most appropriate answer of the following :<br>i) _____ command is used to rename a file in linux environment.<br>-ii) Computers use the _____ language to process data.<br>-iii) Scanner and Mouse are examples of _____ types of devices.<br>-iv) Internet Explorer is an example of _____ software. (System/Application)                                                           | 02    | CO |
| (c)          | Answer the following:<br>-(i) How many bits are there in the following - byte, megabyte, and gigabyte?<br>-(ii) Enlist the name of Linux commands that are used for development of the program using C language.                                                                                                                                                                                  | 03    | CC |
| (d) (i)      | Draw a flowchart to check whether the given integer number is a prime number or not. In flowchart handle all cases of the number i.e. negative number, positive number and non-integer.                                                                                                                                                                                                           | 05    | CC |
| (ii)         | How run-time error is different from compile-time errors?                                                                                                                                                                                                                                                                                                                                         | 02    | CC |
| OR           |                                                                                                                                                                                                                                                                                                                                                                                                   |       |    |
| (e) (i)      | Draw a flowchart to calculate the sum of series.<br>$\text{Sin}(x) = x - x^3/3! + x^5/5! - x^7/7! + \dots$                                                                                                                                                                                                                                                                                        | 05    | CC |
| (ii)         | What is a variable? Explain the ways to declare scope of a variable.                                                                                                                                                                                                                                                                                                                              | 02    | CC |
| Q.2 (a)      | -Differentiate between type casting and type conversion with the help of an example.                                                                                                                                                                                                                                                                                                              | 02    | CO |
| (b)          | Identify the most appropriate answer of the following:<br>i) _____ is the range of values that can be stored by int datatype in C.<br>-ii) _____ is a type of _____ operator.<br>-iii) _____ is the number of whitespace characters allowed in C language.<br>-iv) _____ case is matched in a switch statement, when all cases are unmatched.                                                     | 02    | CO |
| (c)          | Write a program that prints a table with each line giving an integer, its square, and its cube. Ask the user to input the lower and upper limits for the table.                                                                                                                                                                                                                                   | 03    | CO |
| (d)          | Evaluate the expressions for the following, assume that the variables are declared and initialized as follows:<br><code>int x = 3, y = 4, z = 5;</code><br><code>float a = 2.5, b = 3.5, c = 4.5;</code><br>- i) <code>(float)b * (y % x) * y / 2    z * x;</code><br>- ii) <code>++x / 4 * z++ + y / x++ * b;</code><br>- iii) <code>y &lt; c    x &gt; a * ++x &amp;&amp; z &lt;= 5 * b;</code> | 07    | CO |
| OR           |                                                                                                                                                                                                                                                                                                                                                                                                   |       |    |
| (e)          | Write a program to find the difference of two dates in years, month, and days: Find whether the entered dates are valid or-not. Assume that the first date falls before second date.                                                                                                                                                                                                              | 07    | CO |
| Q.3 (a)      | What is an infinite loop? Write the syntax of infinite loop using for loop.                                                                                                                                                                                                                                                                                                                       | 02    | CO |



**SGSITS-CAMPUS-PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

- (b) Explain the output of the following C code with proper justification? 02 CO3 BT4 2.5.2
- ```
#include <stdio.h>
int main(){
    int i, j;
    for (i = 2; i < 10; i++) {
        for (j = 2; j <= 5; j++)
            if (!(i % j))
                break;
        if (j > (i / j))
            printf("%d ", i);
        return 0; }
}
```
- (c) Write the following function in a recursive manner. 03 CO3 BT4 3.8.2
- ```
int function(int n)
{
    int sum=0;
    int count=1;
    while(count <=n) {
        sum = sum+ count;
        count=count+2; }
    return sum; }
```
- (d) Two different numbers are called amicable numbers, if the sum of the proper divisors of each is equal to the other number. For example 220 and 284 are amicable numbers.  
Sum of proper divisors of 220 = 1+2+4+5+10+11+20+22+44+55+110 = 284  
Sum of proper divisors of 284 = 1+2+4+71+142 = 220  
Write a function to print all amicable numbers in a given range.
- OR
- (e) Consider a sorted array X of n elements. Given a key to be searched, the array is partitioned into three halves at the elements mid1 and mid2. mid1 indicates 1/3<sup>rd</sup> index and mid2 indicates 2/3<sup>rd</sup> index of the sorted array. Now comparing the key value to mid1 or mid2 is it determined whether the key lies in first third, middle third or last third of the array. Continue the process until the key is found, if it exists in the array. Write a recursive function TernarySearch(), which takes 4 arguments as input: array, left index, right index, key to be searched. The program should return the value of the key and index, if it exists in the array, otherwise returns -1. 07 CO3 BT3 3.8.2
- Q.4 (a) Explain the output of the following C code with proper justification? 02 CO4 BT4 2.5.2
- ```
#include <stdio.h>
int main(){
    int arr[5] = {1, 2, 3, 4, 5};
    int *ptr1 = arr;
    int *ptr2 = ptr1+3;
    int *ptr3 = ptr2+3;
    printf("%d\n", *ptr2);
    printf("%ld\n", ptr2-ptr1);
    printf("%d\n", *(arr+4));
    return 0; }
```
- (b) Differentiate between call by value and call by reference with the help of an example. 02 CO4 BT2 1.7.1
- (c) Write a program using function reverse (), that takes a string as argument and reverses only a portion of the string using pointers. 03 CO4 BT3 4.5.1
- (d) Consider a 1-D array of integers 0's and 1's. Write a program which reads the 0/1 array having n (<20) elements and then prints the length of the longest sequence of 1's.  
Example, if the input array is int arr[15] = {0,1,0,1,1,0,0,1,1,1,0,1,0,1} the length of the longest run of 1's is 4. 07 CO4 BT3 4.5.1

OR

- (e) A record of "n" employee using an array of structures with four fields (empId, ename, age, salary) is maintained. Each field is of an appropriate data type. 07 CO4 BT3 4.5.1

**SGSITS-CAMPUS PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

of empId and name.

- Q.5 (a) Compare and contrast the structured programming and object oriented programming. 02 CO4 B'
- (b) Explain the visibility of base class members for the access specifiers: private, protected, and public. 02 CO4 B'
- (c) Differentiate between the following: 03 CO4 B'  
(i) Function overloading and Operator overloading  
(ii) Abstraction and Encapsulation
- (d) Create a class named Shape with a function that prints "This is a shape". Create another class named Polygon inheriting the Shape class with the same function that prints "Polygon is a shape". Create two other classes named Rectangle and Triangle having the same function which prints "Rectangle is a polygon" and "Triangle is a polygon" respectively. 07 CO4 B
- OR
- (e) Implement a class 'printdata' with two-member functions all with the same name 'print' 07 CO4 E  
void print(int) – outputs value –, that is, value followed by the value of the integer  
void print (int, int) – outputs value – [, ], that is, value followed by the two integers separated by a comma in square brackets.  
Write a main function that uses the above class and its member functions.

xxxxx



# SGSITS CAMPUS PHOTOCOPY CENTRE

Below Gym Centre, SGSITS Campus

Mobile : 9893555950

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

88001127028

812

JUNE-JULY 2022 EXAMINATION  
1 B.Tech. (4YDC) EXAMINATION  
CO 10504: COMPUTER PROGRAMMING

Time: 3 Hrs.]

[Max. Marks : 70  
[Min Marks : 22

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

NOTE: 1) Answer to the point and make assumptions wherever required and clearly state them. Assume C programming language wherever required. 2) Attempt all parts of same question at one place and each question carries five sub parts a, b, c, d. Part a, b, c is compulsory and attempt any one from d and e.

- |                                                                                                                                                     | CO | BT      | PI    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----|---------|-------|
| Q.1 (a) Explain how negative numbers are represented in the computer memory? Give one example.                                                      | 02 | CO1 BT2 | 1.2.1 |
| (b) Identify the most appropriate answer of the following :                                                                                         | 02 | CO1 BT1 | 1.7.1 |
| i) _____ are the memory of the CPU.                                                                                                                 |    |         |       |
| ii) 2MB is equivalent to _____ bits.                                                                                                                |    |         |       |
| iii) _____ command is used delete the file in Linux operating system.                                                                               |    |         |       |
| iv) 'gedit' is an example of _____ software.                                                                                                        |    |         |       |
| (c) Draw a flowchart to take four points from user (x1,y1), (x2,y2), (x3,y3) and (x4,y4) and find out if these points form an parallelogram or not. | 03 | CO1 BT3 | 2.5.2 |
| (d) Differentiate between the following with the help of an example:                                                                                | 07 | CO1 BT2 | 1.7.1 |
| i) Source code file and executable file                                                                                                             |    |         |       |
| ii) Unary operators and ternary operator                                                                                                            |    |         |       |
| iii) short int and long int                                                                                                                         |    |         |       |
| iv) Type conversion and type casting                                                                                                                |    |         |       |

OR

- |                                 |    |         |       |
|---------------------------------|----|---------|-------|
| (e) Solve the following :       | 07 | CO1 BT3 | 1.2.1 |
| i) $(A245.219)_{13} = (?)_6$    |    |         |       |
| ii) $(F3.4532)_{15} = (?)_8$    |    |         |       |
| iii) $(1B123.CA2)_{19} = (?)_5$ |    |         |       |
| iv) $(1341.567)_9 = (?)_{16}$   |    |         |       |

- |                                                                                                                                                   |    |         |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----|---------|-------|
| Q.2 (a) What is the difference between constant using #define macro and const keyword?                                                            | 02 | CO1 BT2 | 1.7.1 |
| (b) Discuss about iterative statements? What are the different types of iterative statements in C?                                                | 02 | CO2 BT2 | 1.7.1 |
| (c) What is an identifier (variable)? What are the rules to construct identifier (variable)? Classify the following as valid/invalid identifiers. | 03 | CO2 BT2 | 1.7.1 |
| i) num2 ii) \$num1 iii) +add iv) a_2 v) 199_space vi) _apple                                                                                      |    |         |       |
| (d) Evaluate the expressions for the following, assume that the variables are declared and initialized as follows:                                | 07 | CO2 BT5 | 1.2.1 |
| int x = 1, y = 2, z = 3;                                                                                                                          |    |         |       |
| float a = 1.1, b = 2.2, c = 3.3;                                                                                                                  |    |         |       |
| i) $a * z \% y * (\text{int})c / 5 \&\& y > x \parallel y++$ ;                                                                                    |    |         |       |
| ii) $++x / 2 / y++ * z \&\& x++ * a \&\& z$ ;                                                                                                     |    |         |       |
| iii) $(x <= z) \parallel ((++y > b * ++x) \&\& (z >= 5 * \text{int}(c)))$ ;                                                                       |    |         |       |

OR

- |                                                                                                                                                         |    |         |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------|-------|
| (e) Write a program which reads an integer $n > 10$ and a real variable $x$ , $0 <=  x  <= 1$ . Compute and print the sum of the series upto $n$ terms: | 07 | CO3 BT4 | 2.6.3 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------|-------|

$$S_n(x) = x + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1}{2} \cdot \frac{x^5}{5} + \dots + \frac{1}{2} \cdot \frac{x^{2l-1}}{2l-1} + \dots$$

The program should NOT use any standard library function.

- |                                                                                                                                                                                                                              |    |         |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------|-------|
| Q.3 (a) Differentiate between printf(), fprintf(), puts(), and fputs().                                                                                                                                                      | 02 | CO3 BT2 | 2.6.4 |
| (b) Tabulate the scope and life time of the following :                                                                                                                                                                      | 02 | CO3 BT1 | 1.7.1 |
| (i) External variable (ii) Static variable (iii) Automatic variable (iv) Register variable                                                                                                                                   |    |         |       |
| (c) Write a C function isprime(num) that accepts an integer argument and returns 1 if the argument is a prime or 0 otherwise. Write a program that invokes this function to generate prime numbers between the given ranges. | 03 | CO3 BT4 | 4.6.3 |
| (d) Write a recursive program to calculate the value of $nCr$ . $nCr$ is the number of ways of selecting $r$ items given $n$ items. The recurrence for calculating $nCr$ is given by:                                        | 07 | CO3 BT4 | 4.5.1 |
| $nCr = {}^{n-1}C_r + {}^{n-1}C_{r-1}$ . Assume that $n \geq r$ .                                                                                                                                                             |    |         |       |

OR

- |                                                                                                   |    |         |       |
|---------------------------------------------------------------------------------------------------|----|---------|-------|
| (e) Consider the Peasants' Algorithm for multiplication of two positive integers. It works in the | 07 | CO4 BT4 | 4.6.3 |
|---------------------------------------------------------------------------------------------------|----|---------|-------|

**SGSITS-CAMPUS PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

following manner.

- Write the two numbers in two columns.
- Keep updating according to the following procedure until the number in the first (i.e., left) column becomes 1.
- Halve the number in first column (integer division), double the number in the second column.
- At the end, sum up all the numbers in the second column, for which, corresponding number in the first column is odd.

Example of multiplication using Peasants' algorithm:  $13 \times 8: 64+32+8 = 108$ 

|    |    |
|----|----|
| 13 | 8  |
| 6  | 16 |
| 3  | 32 |
| 1  | 64 |

Fill in appropriate blanks in function for calculating multiplication using Peasants' Algorithm.

void peasantMultiplication(int num\_col1, int num\_col2, int answer)

```

{
    int count=0;
    while(1){
        count++;
        if( _____ )
            answer = _____ + _____;
        if( _____ )
            _____;
        num_in_col1 = _____ / 2;
        num_in_col2 = _____ * 2; } }

```

Q.4 (a) Explain the following:

- Write a statement to read a file in append mode.
- Write the purpose of the fseek() function.

02 CO4 BT

(b) Is it possible to return multiple values from a function? Justify the statement with the help of an example.

02 CO3 BT

(c) Write a program that takes string as input from the user and replaces each alphabet in the string with next alphabet except letters "z" "Z" and "a" "A". For example-  
Input string: "Programming in Z is fun" Output String: "Qsphsannjoh jo Z jt gvo"

03 CO4 BT

(d) Write a program to read details of 'n' number of employees from keyboard. The employee record should have three fields (id, name, salary). Print the details of employees whose salary is above 5000 and also store the information in file having name record.txt.

07 CO4 BT

OR

(e) A rectangle has two sides parallel to the x and y axis. Define a structure 'rectangle' to store coordinates of lower left and upper right vertices defined by following structure 'point'.

07 CO3, BT  
CO4

```

typedef struct{
    float x;
    float y;
}point;

```

Write a function:

int inside(point p, struct rectangle r) that returns 1 if the point p is inside or on the rectangle, and 0 otherwise.

Q.5 (a) Differentiate between the class and object. Explain with the help of programming example.

02 CO4 BT

(b) Explain the visibility of base class members for the access specifiers: private, protected, and public.

02 CO4 BT

(c) Explain the concept of polymorphism with the help of programming example.

03 CO4 BT

(d) Differentiate between the following:

07 CO4 BT

- Function overloading and Operator overloading
- Abstraction and Encapsulation
- Static and Dynamic binding
- Multiple Inheritance and Multilevel Inheritance

OR

(e) (i) Enlist the major differences between procedures oriented and object oriented programming language.

03 CO4 BT

(ii) Explain the concept of Inheritance. Demonstrate its use using programming example.

04 CO4 BT

XXXXX



**SGSITS CAMPUS PHOTOCOPY CENTRE****Below Gym Centre, SGSITS Campus****Mobile : 9893555950**

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

323

**NOV-DEC 2022 EXAMINATION  
I B.Tech. (4YDC) EXAMINATION  
CO 10504: COMPUTER PROGRAMMING**

Time: 3 Hrs.]

[Max Marks: 70

**TOTAL NO. OF QUESTIONS IN THIS PAPER: 5****NOTE: 1) Answer to the point and make assumptions wherever required and clearly state them.****2) Attempt all parts of same question at one place and each question carries five sub parts a, b, c, d, e. Part a, b, c is compulsory and attempt any one from d and e.**

| Question No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks | CO       | BT  | PI    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------|-----|-------|
| Q.1          | (a) Differentiate between:<br>(i) Primary and Secondary Memory<br>(ii) Keywords and Identifiers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 02    | CO1      | BT1 | 1.7.1 |
|              | (b) Identify the most appropriate answer of the following :<br>i) _____ command is used to compile a program.<br>ii) Computers use the _____ language to process data.<br>iii) Monitor and Printer are examples of _____ types of devices.<br>iv) MS Word is an example of _____ software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 02    | CO1      | BT1 | 1.7.1 |
|              | (c) Take three points as input (x1, y1), (x2, y2), and (x3, y3). Draw a flow chart to find whether the points form a triangle or not.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 03    | CO1, CO3 | BT3 | 2.5.2 |
|              | (d) Answer the following:<br>(i) Explain the steps involved in solving a problem using computer.<br>(ii) What is the difference between creating constant using #define macro and const keywords?<br>(iii) Write 3 examples of input and output devices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 07    | CO1      | BT2 | 1.7.1 |
|              | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |          |     |       |
|              | (e) Solve the following :<br>(i) $(A212.1029)_{15} = (?)_7$<br>(ii) $(2173.3312)_{11} = (?)_8$<br>(iii) $(B17.258)_{16} = (?)_5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 07    | CO1      | BT3 | 1.2.1 |
| Q.2          | (a) Differentiate between implicit and explicit conversions with the help of an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 02    | CO2      | BT2 | 1.7.1 |
|              | (b) What is meant by control statements? What are the different types of control statements in C languages?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 02    | CO2      | BT2 | 1.7.1 |
|              | (c) Answer the following:<br>(i) What is the purpose of different types of brackets supported by C languages?<br>(ii) What is a variable? Explain the ways to declare scope of a variable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 03    | CO2      | BT2 | 1.7.1 |
|              | (d) Evaluate the expressions for the following, assume that the variables are declared and initialized as follows:<br>int x = 2, y = 3, z = 5;<br>float a = 1.5, b = 2.5, c = 3.5;<br>i) $b * y \% x * (\text{float})y / 2 \parallel z * x$ ;<br>ii) $++x / 2 * z++ + y / x++ * b$ ;<br>iii) $(y < c) \parallel (x > b * x++) \&\& (z < 15 * b)$ ;                                                                                                                                                                                                                                                                                                                                                                                                             | 07    | CO2      | BT5 | 1.2.1 |
|              | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |          |     |       |
|              | (e) Write a program to find the given number is divisible by 2, 3, 5 or 7. If number is divisible by 2, then output is "Number is divisible of 2". If number is divisible by 3, then output is "Number is divisible of 3.", If number is divisible by 5, then output is "Number is divisible of 5.", If number is divisible by 7, then output is "Number is divisible of 7.", If number is divisible by 3 and 7, then output is "Number is divisible of 3. Number is divisible of 7.", If number is divisible by 3, 5 and 7, then output is "Number is multiply of 3: Number is divisible of 5. Number is divisible of 7." ....<br>a. Write the above program using if-else/ff-else-if statements.<br>b. Write the above program using switch/case statements. | 07    | CO2      | BT4 | 2.6.3 |

# SGSITS-CAMPUS-PHOTOCOPY CENTRE

Below Gym Centre, SGSITS Campus

Mobile : 9893555950

Dear Students,

Please do not share papers with your friends by making the pdf.  
Otherwise we would be helpless in next sem for providing the paper updates.

|     |     |                                                                                                                                                                                                                                                                                                              |    |     |
|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
| Q.3 | (a) | What is an infinite loop? Write the syntax of infinite loop using for loop.                                                                                                                                                                                                                                  | 02 | CO  |
|     | (b) | What will be the output of the following program, when program is executed? Justify your answer.<br>#include<stdio.h><br>int main() {<br>int i, sum1 = 0, sum2 = 0;<br>for (i = 1; i<=11; i++)<br>if((i*i*i*2+3)>=(i*i*4+5))<br>sum1++;<br>else<br>sum2++;<br>printf("%d %d", sum1, sum2);<br>return 0;<br>} | 02 | CO  |
|     | (c) | Write a program to enter a number and find the reverse of that number. Also display the double of the reverse number. Do not use arrays.                                                                                                                                                                     | 03 | CO3 |
|     | (d) | Write a recursive function (function name: binomial) to find the value of Binomial Coefficient ${}^nC_r$ where the boundary conditions are ${}^nC_r = 1$ if $r=0$ and ${}^nC_r = 1$ if $r=n$ .                                                                                                               | 07 | CO3 |
|     |     | OR                                                                                                                                                                                                                                                                                                           |    |     |
|     | (e) | Write a program using function that take the number parameter as an integer from user and convert and print the number of days, hours and minutes. (i.e. if number = 63 then the output should be 0:1:3). Separate the number of days, hours and minutes with a colon.                                       | 07 | CO3 |
| Q.4 | (a) | What is the difference between the following declarations?<br>int arr [20]; int* arr1[20]; int (*arr2)[20]; int **arr3[20];                                                                                                                                                                                  | 02 | CO4 |
|     | (b) | Differentiate between call by value and call by reference with the help of an example.                                                                                                                                                                                                                       | 02 | CO4 |
|     | (c) | Write a program to count the number of occurrences of any two vowels in succession in a line of text. For example, in the sentence "Please read this application and give me gratuity" example of such occurrences are ea, ea, ui.                                                                           | 03 | CO4 |
|     | (d) | Write a C program to maintain a record of "n" student details using an array of structures with four fields (Roll number, Name, Marks, and Grade). Each field is of an appropriate data type. Print the marks of the student given student name as input.                                                    | 07 | CO4 |
|     |     | OR                                                                                                                                                                                                                                                                                                           |    |     |
|     | (e) | Given a text file, write a program to create another text file deleting the words "a", "the", "an" and replacing each one of them with a blank space.                                                                                                                                                        | 07 | CO4 |
| Q.5 | (a) | Describe an object and a class? Write a program which shows how to define a class and object.                                                                                                                                                                                                                | 02 | CO4 |
|     | (b) | Explain the visibility of base class members for the access specifiers: private, protected, and public.                                                                                                                                                                                                      | 02 | CO4 |
|     | (c) | What is meant by inheritance? Explain with examples, the various types of inheritance supported in C++.                                                                                                                                                                                                      | 03 | CO4 |
|     | (d) | Differentiate between the following:<br>(i) Function overloading and Operator overloading<br>(ii) Abstraction and Encapsulation<br>(iii) Static and Dynamic binding                                                                                                                                          | 07 | CO4 |
|     |     | OR                                                                                                                                                                                                                                                                                                           |    |     |
|     | (e) | (i) Differentiate between procedural programming and object oriented programming.                                                                                                                                                                                                                            | 04 | CO4 |
|     |     | (ii) Differentiate between compile time and run time polymorphism.                                                                                                                                                                                                                                           | 03 | CO4 |
|     |     | XXXXX                                                                                                                                                                                                                                                                                                        |    |     |



Online Feb. 2021 EXAMINATION  
I B.E./ B.Tech (4YDC) EXAMINATION  
CO 10504: COMPUTER PROGRAMMING

Time: 90 Mins.]

[Max. Marks: 40

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

**NOTE:**

- 1) Answer to the point and make assumptions wherever required and clearly state them.
- 2) Attempt all parts of same question at one place and each question carries two sub parts a and b. Attempt any one from a and b.

| Question No. | Question                                                                                                                                                                                                                                                                                                                                                                                                 | Marks | CO       | BL  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------|-----|
| Q.1 (a) (i)  | Solve the following:<br>i) $(1834.342)_{11} = (?)_9$<br>ii) $(C3F.C4B)_{17} = (?)_5$<br>iii) $(647.123)_8 = (?)_2$                                                                                                                                                                                                                                                                                       | 04    | CO1      | BL2 |
|              | (ii) Draw a flowchart to print the sum of square of all even numbers between 2 to 500 (both numbers inclusive).                                                                                                                                                                                                                                                                                          | 04    | CO1      | BL2 |
|              | OR                                                                                                                                                                                                                                                                                                                                                                                                       |       |          |     |
| (b) (i)      | Solve the following:<br>i) $(686.3A2)_{16} = (?)_8$<br>ii) $(324.3124)_{11} = (?)_7$<br>iii) $(231.212)_5 = (?)_9$                                                                                                                                                                                                                                                                                       | 04    | CO1      | BL3 |
|              | (ii) Draw a flowchart to find if the given number is a palindromic prime number or not.                                                                                                                                                                                                                                                                                                                  | 04    | CO1      | BL2 |
| Q.2 (a) (i)  | Solve the following for the values int a = 10, b = -15, c = 20, d = -6, Z = 0;<br>1) $Z = a < b \parallel ++c * 2 \% 2 + 1 \&\& 2 + !b \% 2$ ;<br>2) $Z = ++b * d > a \% b * -1 ? a \% 2 - 20 : c * 10 + 5$ ;<br>3) $Z = a + 5 * c * d < a \&\& b$                                                                                                                                                       | 04    | CO1      | BL5 |
|              | (ii) An astrologist map large numbers to single digit number between 1 and 9 in order to tell your future. Suppose Ram went to an astrologist to find his future. Suppose he tell his date of birth as 19011987 and astrologist maps it to 9. Write a program, which takes a number as input and it reduces it to a single digit.                                                                        | 04    | CO2      | BL3 |
|              | OR                                                                                                                                                                                                                                                                                                                                                                                                       |       |          |     |
| (b) (i)      | Write a program using switch to take a 4-digit integer as input and check if it is divisible by 2, 3, 4, and 12.                                                                                                                                                                                                                                                                                         | 04    | CO2, CO3 | BL4 |
|              | (ii) You are given a sequence of non-negative integers terminated by -1. You have to output 1 if there are atleast 2 distinct elements in the sequence and 0 if the sequence consists of only 1 integer. Note that -1 is not part of the sequence. The sequence is not necessarily sorted. Note: Don't use arrays to this question.<br>Example: Input: 1 1 1 2 -1 Output: 1, Input: 4 4 4 4 -1 Output: 0 | 04    | CO3      | BL4 |

|     |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |     |
|-----|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
| Q.3 | (a) | (i)  | Write a program to input two 1-D int type arrays A[10] and B[10] which contain positive integers and store the set difference A-B in C. Fill the remaining positions of C[10] with -1. Assume that all the elements in A and B are distinct.<br>For Example: Set difference A-B is the subset of A which have elements which are not in B.<br>A={1,2,6,11,3}, B={2,7,9,10,11} then A-B={1,6,3}.                                                                             | 04 | CO  |
|     |     | (ii) | Write a program that takes two positive integers n1 and n2 from the user and check if they are permutation of each other.                                                                                                                                                                                                                                                                                                                                                   | 04 | CO  |
|     |     |      | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |     |
|     | (b) | (i)  | The following program intends to compute $\pi$ by Newton's formula upto the 10000-th term:<br>$\frac{\pi}{2} = \sum_{k=0}^{\infty} \frac{2^k (k!)^2}{(2k+1)!}$<br>Fill up the dashed lines in the program<br>#include <stdio.h><br>int main ()<br>{<br>double sum = ____;<br>double term = ____;<br>int k = ____;<br>for(; k < 10000; ++k) {<br>term = ____;<br>sum = ____;<br>}<br>sum = ____;<br>printf("%lf", sum);<br>return 0;<br>}                                    | 04 | CO  |
|     |     | (ii) | Write a recursive function for finding length of strings and display the string the string in front and reverse order.                                                                                                                                                                                                                                                                                                                                                      | 04 | CO3 |
| Q.4 | (a) | (i)  | Write a Program using pointers to find whether the two strings are anagram or not. For Example: LISTEN and SILENT are two anagram strings.                                                                                                                                                                                                                                                                                                                                  | 04 | CO3 |
|     |     | (ii) | Given the current time and time required for a class, it is desired to find out the time when the class ends. Write a structure to represent time in hour, minutes, and seconds. Also write a function which gets the current time and class time, and returns the end time of class.                                                                                                                                                                                       | 04 | CO3 |
|     |     |      | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |     |
|     | (b) | (i)  | The absolute distance between two integers x1 and x2 is given by  x2-x1 . Write a function which sorts an array x[] of n integers in ascending order of their absolute distances with a given number z. For example, given x[] = {9, 1, 12, 4, 2} and z=6, the sorted array will be x[] = {4, 9, 2, 1, 12}. Note that 4 is closest to 6, and 12 is farthest from 6 in terms of absolute distance. The function Prototype will be:<br>void sort_abs (int x[], int n, int z); | 04 | CO4 |



|     |     |                                                                                                                                                                                                                                                 |    |     |          |
|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----------|
|     |     | (ii) Write a program which uses a structure that read roll number, name, and percentage of 5 students. If the percentage of the student is greater than 70 then write its information to the file 'F1.txt' otherwise write it to file 'F2.txt'. | 04 | CO4 | BL3      |
|     |     |                                                                                                                                                                                                                                                 |    |     |          |
| Q.5 | (a) | (i) Enlist the major differences between procedures oriented and object oriented programming languages.                                                                                                                                         | 04 | CO4 | BL4      |
|     |     | (ii) Differentiate between the following with programming example:<br>i) Function overloading and Operator overloading<br>ii) Object and Class<br>iii) Abstraction and Encapsulation                                                            | 04 | CO4 | BL2, BL3 |
|     |     | <b>OR</b>                                                                                                                                                                                                                                       |    |     |          |
|     | (b) | (i) Differentiate between the following with programming example:<br>i) Overloading and overriding<br>ii) Multiple Inheritance and multilevel inheritance<br>iii) Static binding and dynamic binding                                            | 04 | CO4 | BL2, BL3 |
|     |     | (ii) Enlist the major advantages of C++ programming language. Justify your answer using a programming example.                                                                                                                                  | 04 | CO4 | BL4      |
|     |     | XXXXX                                                                                                                                                                                                                                           |    |     |          |

12/12/19

~~CR-1001-001104~~

NOV-DEC 2019 EXAMINATION  
I B.E. (4YDC) EXAM  
CO10504: COMPUTER PROGRAMMING

Time: 3 Hrs.]

[Max  
[Min. Pas

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note: Attempt all the questions. Each question has four sub-parts a, b, c and d. Attempt any of (a) and (b), and similarly any one part out of (c) and (d). Attempt all sub-parts of a question

Q 1. (a) Write/fill the most appropriate answer of the following :

[05]

- i) Surgeons can perform delicate operations by manipulating devices through computers instead of manually. This technology is known as \_\_\_\_\_.
- ii) A hexadecimal number that comes immediately after a hexadecimal number 5507F is \_\_\_\_\_.
- iii) The term bit is short for \_\_\_\_\_.
- iv) Computers use the \_\_\_\_\_ language to process data.
- v) A \_\_\_\_\_ is approximately a ten lacs bytes.

OR

(b) Write/fill the most appropriate answer of the following :

[05]

- i) The name for the way that computers manipulate data into information is called \_\_\_\_\_.
- ii) A hexadecimal number that comes just before the hexadecimal number A0F0 is \_\_\_\_\_.
- iii) A \_\_\_\_\_ is approximately one thousand bytes.
- iv) Apple Macintoshes (Macs) and windows are two different \_\_\_\_\_.
- v) Word processing, spreadsheet, and photo-editing are examples of \_\_\_\_\_.

(c) i) Draw a flowchart that accepts a year between 1900 and 2100 and prints the day of the week for 1<sup>st</sup> Jan of that year. Let the day on the 1<sup>st</sup> Jan 1900 is Monday. [05]

ii) Solve the following :

[04]

$$(1212.2019)_{12} = (?)_7$$

OR

(d) i) Draw a flowchart to accept an English string and an integer n and shift each letter by n. See examples below. [05]

Input: Enter the string: APPLE

Enter the shift value: 2

Output: CRRNG

Input: Enter the string: APPLE

Enter the shift value: -1

Output: ZOOKD

ii) Solve the following :

[04]

$$(2010.1212)_7 = (?)_{12}$$



Q 2. (a) Evaluate the expressions for the following, assume that the variables are declared and initialized as follows: [06] CO4 BL3

int x = 1, y = 2, z = 4;

float a = 0.1, b = 0.2, c = 0.3;

- i)  $z + y++ - a * (\text{int})c / 3 / z$  ;
- ii)  $++z \% y * x++ * 7 / x - a$ ;
- iii)  $x / y / b / c / z / a$  ;

OR

(b) i) From the C-program expression  $((((-x / 1) + 44) * (x - 3) \parallel (x - 2)) > (x * (2 / z)) \&\& ((z * 4) > 10)))$ , remove all the unnecessary parenthesis so that the meaning does not change. [06] CO4 BL3

ii) Bracket the C-program expression  $x * 3 < 10 \parallel -z * y / 4 \&\& z + y - 4 \% 2$  so as to show in which order the compiler evaluates the expression.

(c) iii) What is variable? Explain the ways to declare scope of a variable. [02] CO4 BL3

iv) What is meant by iterative statements? What are the different types of iterative statements in C? [03] CO3

v) Given an array of N positive integers, where  $N > 10$ , we want to find the top five largest elements. To get you started, here is the main function. You need to write the code for the insert function that inserts an element into an array, topv that is supposed to contain the top five largest elements. [03]

```
int numbers[N];
int main() {
    int i; int topv[5];
    for (i=0; i < 5; i++) topv[i] = -1;
    for (i=0; i < N; i++) insert(topv, numbers[i]);
    for (i=0; i < 5; i++) printf ("%d\n", topv[i]);
}

void insert (int topv[5], int num) {
    /* Write your code here */
}
```

OR

(d) i) What is the difference between constant using #define macro and const keyword? [02] CO4 BL2 CO3 BL3

ii) Is it possible to return multiple values from a function? Justify the statement with the help of an example. [03]

iii) Complete the following program to express a number in terms of its prime factors and their exponents. For example the number 24 can be expressed as  $2^3 * 3^1$ . In general a number(N) can be represented as:  $N = p_1^{e1} * p_2^{e2} * \dots$ . Here,  $p_1, p_2, \dots$  are prime numbers. Note that all the prime numbers and exponents ( $e1; e2; \dots$ ) are greater than 1. Now, complete the following program to print the prime factorization of a number. For example, for 24, you should print  $2^{**}3^{**}3^{**}1$ , where \*\* is the exponential operator. [03]

```
#define NUM_PRIMES 10000
/* primes is an array with the first 10,000 prime numbers */
int primes[] = {2, 3, 5, 7, 11, 13, 17, 19, ... };
```

```

void print_prime_factorization(int n);
void main() {
    int n;
    /* Assumption: n > 1, and all of its prime factors are elements of
    the primes array */
    scanf("%d", &n);
    print_prime_factorization(n);
}

void print_prime_factorization(int n) {

    /* Write your code here */
}

```

- Q 3. (a) i) What is the use of call by value and call by reference? [06]  
 ii) Differentiate between the warning and error at compilation time.  
 iii) What is enumerated data types? Explain with example.

OR

- (b) Write a program/function that takes array of integers of size  $n$  as input and print the sum of all the prime numbers in given input array. For example if input array is [2, 4, 6, 9, 11] then output is 13. [06]  
 (c) Write a program that takes array of integer of size  $n$  as input and print True if the absolute difference between each adjacent pair of elements strictly in increase of order. For example if input array is [1, 3, 7, 2, 9] then output is True (absolute differences between adjacent elements are  $3-1=2$ ,  $7-3=4$ ,  $7-2=5$ ,  $9-2=7$ , and 2, 4, 5 and 7 are in increases order), if input array is [1, 3, 7, 10] then output is False (absolute differences between adjacent elements are  $3-1=2$ ,  $7-3=4$ ,  $10-7=3$  and 2, 4 and 3 are not in increases order). [08]

OR

- (d) Write a program that takes array of integer of size  $n$  as input and print True if the absolute difference between each adjacent pair of elements alternates between increasing strictly and decreasing strictly. For example if input array is [1, 5, 2, 8, 3] then output is True (absolute differences between adjacent elements are  $5-1=4$ ,  $5-2=3$ ,  $8-2=6$ ,  $8-3=5$ , so the differences decrease, increase and then decrease.), if input array is [1, 5, 2, 8, 1] then output is False (absolute differences between adjacent elements are  $5-1=4$ ,  $5-2=3$ ,  $8-2=6$ ,  $8-1=7$ , so the differences decrease, increase and then increase again.). [08]

- Q 4. (a) i) What is pointer variable? When should we use pointers in a C program? [08]  
 ii) What is the difference between the following declarations?

`int* arr1[8]; int (*arr2)[8]; int *(arr3[8]);`

- iii) What is multi-dimensional array? How they are store in the memory?  
 iv) What is the return value of the following function, if this function is called as `fun(43210)`? Also write the meaning of the function in general.

```

int fun(int x){
    int m = 1;
    int a = 0;
    while ( m <= x ){
        m = m*2 ;
        printf("%d\n",m);
        a = a+1 ;
    }
}

```



```

    }
    return(a);
}

```

OR

- (b) i) Write a situation where an infinite loop can be used? Write the syntax of the infinite loop using for and while loop. [08] CO4 BL2 BL4 BL5
- ii) What is the difference between arrays and pointer? Why pointers are not supported by new programming languages?
- iii) Are a[5] and 5[a] same? Justify your answer.
- iv) Consider the following function fun(). Write the meaning of the function fun() in general. For what values of m, function fun() will not terminate?

```

int fun(m) {
    if (m == 0)
        return(0);
    else
        return(m+fun(m-1));
}

```

- (c) i) Define a structure 'Address Book' containing serial number, name, mobile number, e-mail-id and address. [02] CO2 BL3
- ii) Write a program to read the 'Address book' information (serial number, name, mobile number, email-id and address) from the given file. [04]

OR

- (d) i) What is a structure? What is the benefit offered by using a structure over multiple arrays? [06] CO2 BL3
- ii) What is a FILE? What are the different methods to open a file? Explain each with the help of a C Program.

Q 5. (a) What are the key features of C programming language? [06] CO1 BL1

OR

(b) What are the key features of C++ programming language? [06] CO1 BL1

(c) Differentiate between the following: [08] CO1 BL4

- Function overloading and Operator overloading
- Object and Class
- Abstraction and Encapsulation
- Static and Dynamic binding

OR

(d) Differentiate between the following: [08] CO1 BL4

- Overloading and Overriding
- Structure and a Class
- Constructor and Destructor?
- Early and Late Binding

Computer Engineering  
**April-May 2019 EXAMINATION**  
**I B.E. (4YDC) EXAM**  
**CO10504: COMPUTER PROGRAMMING**

Time: 3 Hrs.]

[Max. Marks: 100]  
[Min. Pass: 30]

**TOTAL NO. OF QUESTIONS IN THIS PAPER : 5**

**Note:** Attempt all the questions. Each question has four sub-parts a, b, c and d. Attempt any one of (a) and (b), and similarly any one part out of (c) and (d). Attempt all sub-parts of a question at

- Q 1.(a)** i) List the three significant limitations of the computer.  
ii) List the three factors that not affect the performance of computer.  
iii) Rearrange the following memories in increasing order of data access speed: Register, RAM, Hard disk, Magnetic Tape, Cache memory.  
iv) Explain the working of the keyboard.  
v) Show the octal, binary and hexadecimal equivalents of the decimal number 12.  
vi) List the key features of the high level programming languages.

**OR**

- (b)** i) List the three significant advantages of the computer.  
ii) List the three factors that affect the performance of computer.  
iii) How many bits are there in the following - byte, megabyte, petabyte and gigabyte?  
iv) Explain the working of the mouse.  
v) Show the octal, decimal and hexadecimal equivalents of the binary number 1001001111.  
vi) List the key features of the low level programming languages.

- (c)** i) Draw a flowchart to take input N as a number from user and find and display the sum of its digits.  
ii) If 64 (in base 'x' number system) is equal to 144 (in base 'y' number system), find the possible values of x and y.

**OR**

- (d)** i) Draw a flowchart to calculate and display the simple interest and compound interest for the given appropriate inputs.

- ii) Solve the following:  
 $(AABB.CCDD)_{17} = (X?)_7$

- Q 2.(a)** i) From the C-program expression  $(((-2/((4 * x) + 1) * 3)) > ((x + 2) * (-z))) \&\& ((z - 10))$ , remove all the unnecessary parenthesis so that the meaning does not change.  
ii) Evaluate the expressions for the following, assume that the variables are declared and initialized as follows:

int x = 11, y = 22, z = 33;

float a = 11.11, b = 22.22, c = 33.33;

- a)  $a + b * c * x + 2 - y + 3$ ;  
b)  $y \% 3 * x - * 6 + a - a * b$ ;  
c)  $a * x * b * y / (int)c / z$ ;

**OR**



(b) i) Bracket the C-program expression  $x * x < 10 * c - 2 \parallel z * a + 3 \&\& z * y + 4 \% 2$  so as to show in which order the compiler evaluates the expression. [06]

ii) Evaluate the expressions for the following, assume that the variables are declared and initialized as follows:

int x = 100, y = 10, z = 1;

float a = 100.5, b = 10.5, c = 1.5;

a)  $x < a/3 \&\& x > b/x \parallel c > .01 * c$ ;

b)  $x * (int)a + (int)a * y / c + x * z$ ;

c)  $x / 4 * x + 3 * c - y * x$ ;

(c) i) Write the expression of the each of the following operators? [02]

a) Unary Operators

b) Ternary Operators

ii) Write conditional statements to perform the following operations: [06]

a) Given two numbers, calculate and print their sum if both numbers are either odd or even; otherwise calculate their difference.

b) Determine and print whether the sum of digits of a given three-digit number is odd or even.

c) Print the given single digit number in the text form; for example if input is '3' then output will be 'Three'.

OR

(d) i) Differentiate between the following: [08]

a) Float and double

b) Short and integer

iii) Write looping statements to perform the following operations:

a) Calculate and print the sum of numbers from n to 2n, where n is given.

b) Print uppercase and lowercase letters, as in AaBbCc...Zz.

c) Print the table of the given number n, for example if given number is 2 the output will be:

2x1=2

2x2=4

...

...

2x10=20

Q 3.(a) An Armstrong number is the one in which the sum of cubes of its digit is equal to the number itself. For example: 371 is an Armstrong Number ( $27+343+1=371$ ). Write a function which return 1 if a given number is an Armstrong number and 0 otherwise, to check whether a given number is an Armstrong number or not. Use this function to find the smallest an Armstrong number, greater than a given number N. [07]

OR

(b) A perfect number is a positive integer that is the sum of its proper positive divisors, that is, the sum of the positive divisors excluding the number itself. The first perfect number is 6, because 1, 2, and 3 are its proper positive divisors, and  $1 + 2 + 3 = 6$ . The next perfect number is  $28 = 1 + 2 + 4 + 7 + 14$ . Write a function which a return 1 if a given number is a perfect number and 0 otherwise. Use this function to find the smallest perfect number, greater than a given number N. [07]

- (c) Write a program to print an upper triangular square matrix (for the given size of n) in the following form. For example, if given input matrix size is 4, then output will be:

```

1 5 8 10
0 2 6 9
0 0 3 7
0 0 0 4

```

OR

- (d) Write a program to print a lower triangular square matrix (for the given size of n) in the following form. For example, if given input matrix size is 4, then output will be:

```

1 0 0 0
5 2 0 0
8 6 3 0
10 9 7 4

```

- Q 4.(a) i) Write a function that accept an array of integers and return the pointer to its largest element. [0]

ii) What precaution should be taken while returning a pointer from a function?

iii) Write the following function in a recursive manner.

```

int function(int n)
{
    int sum=0;
    int count=1;
    while(count <= n)
    {
        sum = sum+count;
        count++;
    }
    return sum;
}

```

OR

- (b) i) Give the declarations for the following: [0]

a) Pointer to an array of double.

b) Function which return a pointer to char.

c) Array of pointers to functions with return value unsigned int.

ii) What is the call-by-value and call-by-reference? Explain by some coding example.

iii) Write the following function in an iterative manner.

```

int fun(int a, int b)
{
    while (a != b)
    {
        if (a > b)
        {
            return fun(a - b, b);
        }
        else
        {
            return fun(a, b - a);
        }
    }
    return a;
}

```